What Sparse Autoencoders Miss Is Task-Specific

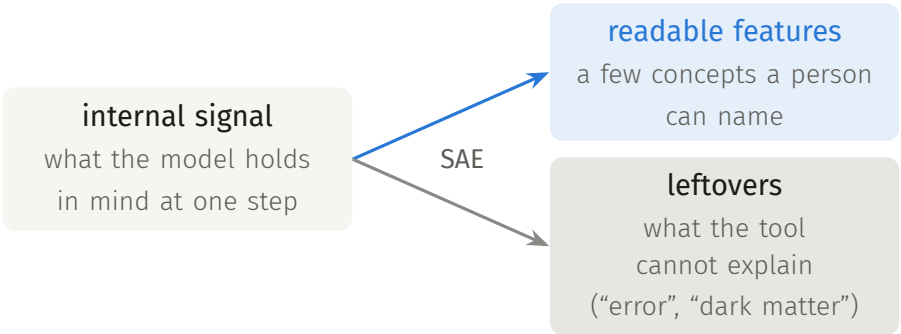
Mapping the blind spot across tasks and languages

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[Course name]

How we look inside a language model



Sparse autoencoder (SAE): the main tool researchers use to read a model’s internal signal [1, 2].

The idea in one slide

WE STARTED WITH

The leftovers hide what the model keeps track of.

→

THE PILOT SHOWED

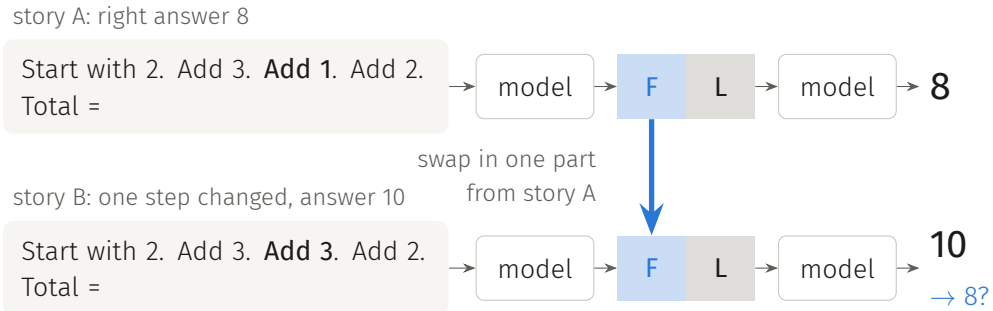
Mostly not. But what the SAE *does* miss is specific to the task.

→

WE PROPOSE

Map this blind spot across more tasks and across Indian languages.

The swap test: does the right answer come back?



F = readable features, L = leftovers. Score: **how much of the right answer comes back**, from 0% (still 10) to 100% (back to 8). Swapping everything gives 100%.

Pilot: four puzzles the model can solve

Running sum

`Start with 2. Add 3. Add 1. Add 2. Total =` 8

keeps a total

Object tracking

`The key is in box B. … The key moves to box C. … The key is in box` C

links objects to boxes

Variable copying

`t = 7. q = 4. x = q. q = 2. x =` 4

follows an assignment

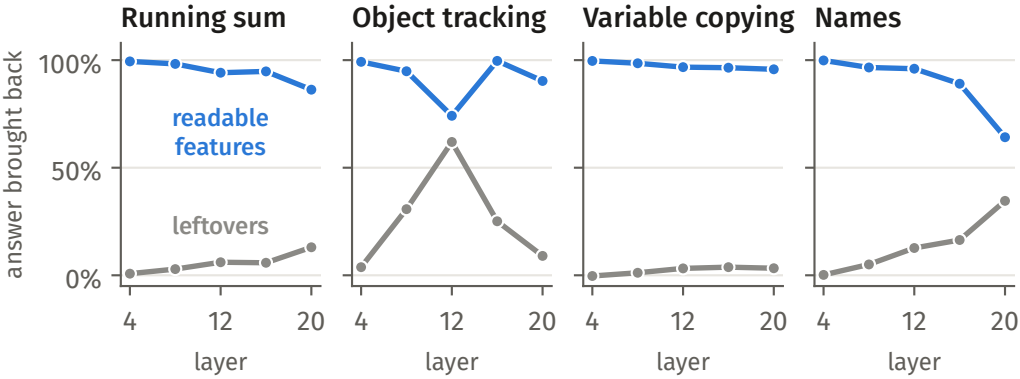
Names (control)

`When Mary and John went to the store, John gave a drink to` Mary

a well-studied case [5]

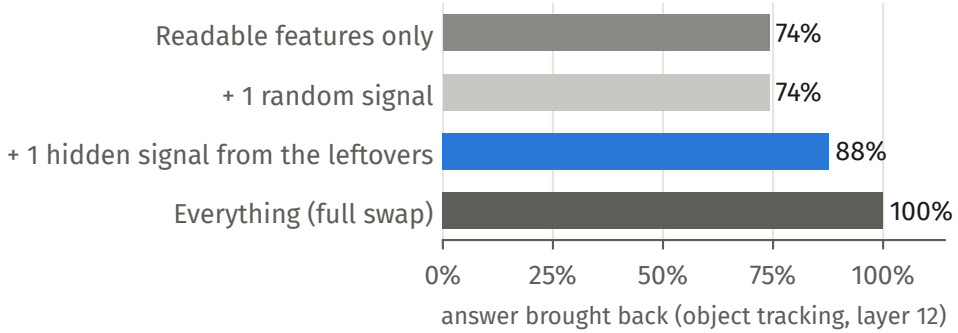
Model: gemma-2-2b (Google) · SAEs: Gemma Scope [4] · 150 test pairs per puzzle · three harder puzzles were dropped: the model could not solve them.

Readable features carry most of the answer



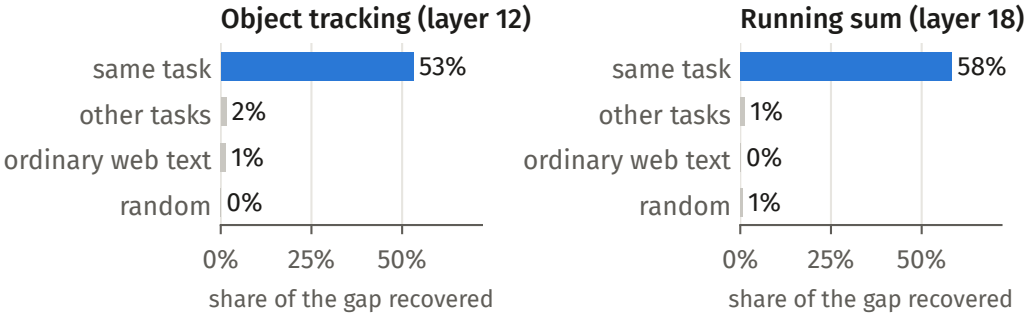
The “leftovers hide it” idea is mostly wrong here. One exception stands out: object tracking, middle layers.

Object tracking leaks: one hidden signal recovers half of it



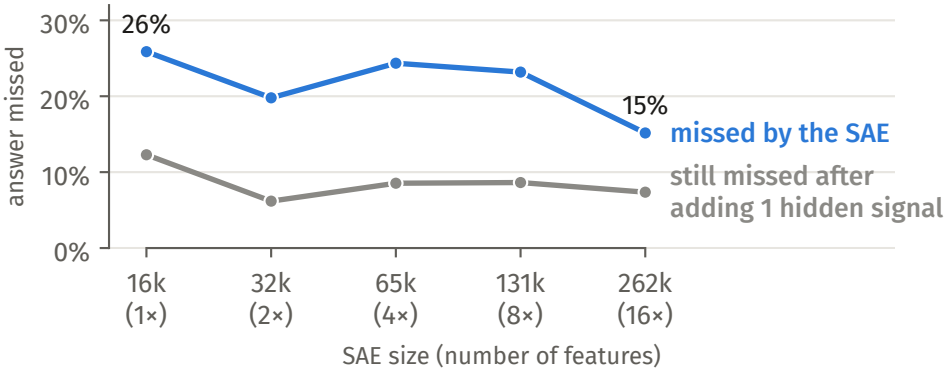
The SAE misses 26% of the answer. A single signal pulled from the leftovers brings 14 points back.

The missing piece is specific to the task



Signals learned from the same task fill the gap. Signals from other tasks or ordinary web text do almost nothing.

A 16× bigger SAE shrinks the gap but does not close it



At every size, one hidden signal still recovers about half of what is missed.

Three ways to fool yourself (we hit all three)

Swap the whole sentence, not just the last word

47% vs 100%

object tracking, layer 12: last word only vs from the first changed word

Check the words don't give the answer away

100%

a reader that sees only the words, no model, already gets variable copying and names right

Readable is not the same as used

84% vs 47%

the box can be read from the leftovers, yet swapping there brings back at most 47%

Next (1): map the effect across tasks

Kind of task	Examples	Status
Linking things	objects in boxes; people and their cities (new)	pilot + new
Keeping a count	running sum; longer chains (new)	pilot + new
Following references	variable copying	pilot
Control	names	pilot

For every task: **how much** does the SAE miss, **is it task-specific**, does a **bigger SAE** fix it?

Binding test: is the leak the model's way of linking "what goes with what" [3]? Change only the hidden signal and see if the model names the other box.

Next (2): does the SAE miss more in Indian languages?

English (pilot baseline)

Hindi

Hinglish (Hindi, Latin script)

Tamil or Bengali

French (non-Indian control)

The SAEs were trained on mostly English text [4].

Three questions

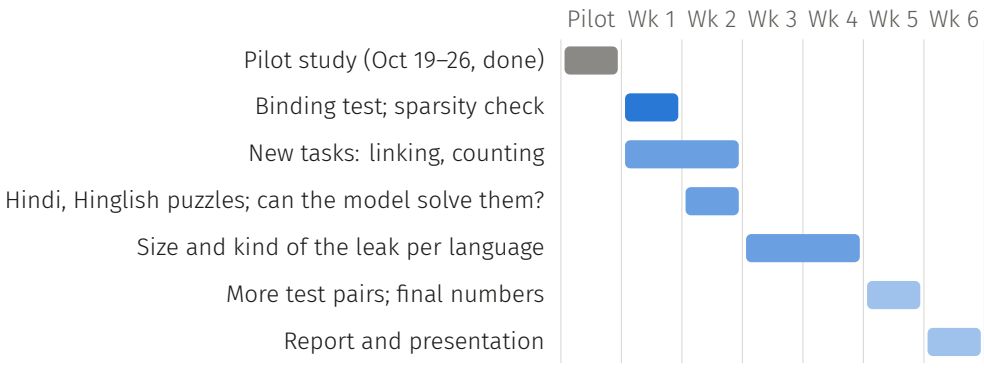
1. Does the SAE explain Hindi text as well as English?
2. Is the leak bigger in Hindi?
3. Is it the same hidden signal? (learn it in English, test it in Hindi)

Same puzzles, translated (illustrative, to be checked)

शुरुआत 2 से करें। 3 जोड़ें। 1 जोड़ें। 2 जोड़ें। कुल = 8

`Jab Rahul aur Priya market gaye, Priya ne ek book di. Book kisko mili? Answer:` Rahul

Plan: six weeks



Takeaways

1. **SAEs keep most of what a model tracks.** Readable features bring back 74% to 100% of the answer.

2. **What they miss is task-specific.** For object tracking, one hidden signal holds half the gap; signals from other tasks or ordinary text do not help, and a 16× bigger SAE does not remove it.

3. **Next: map this blind spot** across more tasks and in Hindi and Hinglish.

Backup: the original claims and what the pilot found

Original claim	Pilot verdict
The leftovers carry what the model tracks	**Mostly no:** readable features bring back 74% to 100%; leftovers 0% to 13% (object tracking up to 62%)
The tracked state sits in the leftovers in a simple, readable shape	**Readable, not used:** a reader decodes it there (84%), but the model does not use it at that spot yet
A few hidden signals in the leftovers fill the gap	**Yes, narrowly:** one signal recovers about half of object tracking's 26% gap; only for that task

Backup: caveats

- One model (gemma-2-2b), one SAE family (Gemma Scope), made-up puzzles.
- The names control also loses 24% to 36% in late layers, but that loss is spread over about 32 signals, so part of the late-layer loss is a general SAE weakness.
- Object tracking's "leftovers only" score (62% at layer 12) is inflated: replacing the leftovers with noise already gives 51%.
- The 131k and 262k SAEs also use more features per word (124 to 139 vs 75 to 89 at 16k), so part of the shrinking gap may come from that, not from size.
- The binding explanation and everything multilingual are plans, not results.

References i

[1] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features/index.html. Published October 4, 2023.

[2] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023. URL https://arxiv.org/abs/2309.08600.

[3] Jiahai Feng and Jacob Steinhardt. How do language models bind entities in context? In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=zb3b6oKO77.

[4] Tom Lieberum, Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma Scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.19. URL https://aclanthology.org/2024.blackboxnlp-1.19/.

[5] Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=NpsVSN6o4ul.